

Task 4 - Requirement Traceability Matrix

Project name: Member Registration & Authentication System

Date: October 5th, 2025

Requirement Collection Approach

To ensure that our system meets the requirements needed, the following approaches will be used for requirements gathering:

1. Interview with stakeholder
 - Conduct interviews with the Springfield Public Library management team, stakeholder and staff to understand the system's functionality and the functional requirements for the project.
 - To understand system's goals and expectations and user experience
2. prototyping
 - develop an early-stage prototype of the system to gather feedback from stakeholders and users
 - plan changes and collect recommendations based on the feedback
3. Review Library's documentations
 - Review existing documentation of the library system to understand the system well and collect necessary information to integrate into our project.
4. Workshops and meetings
 - Organize workshops and meeting with stakeholders and the development team and security specialists to brainstorm and discuss requirements gathered

Requirement Traceability Matrix

Requirement No.	Name	Category	Source	Status
R1	Online client registration form	Functional	- Planning with stakeholders through interview and meetings - system design planning with stakeholder and developer team	Completed
R2	User login and authentication	Functional	- Feedback from users during workshops - Team discussion with developers on system flow - Security team's suggestion and planning	Completed

R3	User must be able to reset the password.	functional	- Stakeholder's suggestions - Reviewing of existing documentation of the system - <u>Final Update</u>: dropped from final scope	Not implemented
R4	System must be available 24/7	Non-functional	- Team discussion and stakeholder meeting	Planned → Supported by Firebase
R5	Secure client's information	Non-functional	- Reviewing system's security policy - Consultation with security analysts - <u>Final Update</u>: PINs currently stored without hashing.	Partially Achieved
R6	System must generate an 8-digit unique Library Card Number	Functional	Development team decision	Completed
R7	Validate input before registration/Login	Functional	Developer implementation and testing	Completed